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Metaheuristic Deep Learning Framework for Automated Leukemia Classification and Severity Grading

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2025

INTRODUCTION

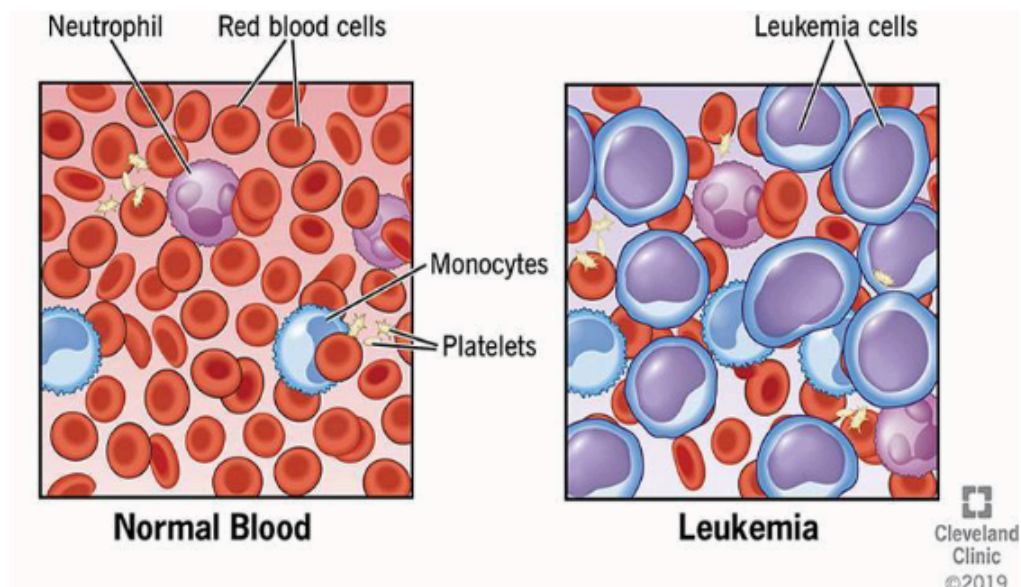
Background of the Problem

Leukemia is a dangerous blood cancer that originates in the bone marrow and severely disrupts the normal production of blood cells. The uncontrolled proliferation of abnormal white blood cells suppresses the formation of healthy red blood cells, platelets, and functional white blood cells. This results in clinical conditions such as anemia, recurrent infections, fatigue, and excessive bleeding or bruising. Certain forms of leukemia progress rapidly, making early detection, accurate classification, and proper grading essential for effective treatment planning and improved patient survival rates. However, the complexity of leukemia and the morphological similarities between different leukemia subtypes make diagnosis challenging, particularly during the early stages of the disease.

At present, leukemia diagnosis primarily depends on the manual examination of peripheral blood smear images by experienced hematopathologists, along with supporting laboratory and clinical investigations. Although these conventional diagnostic methods are widely accepted, they are time-consuming, labor-intensive, and highly dependent on expert judgment, which may vary between observers.

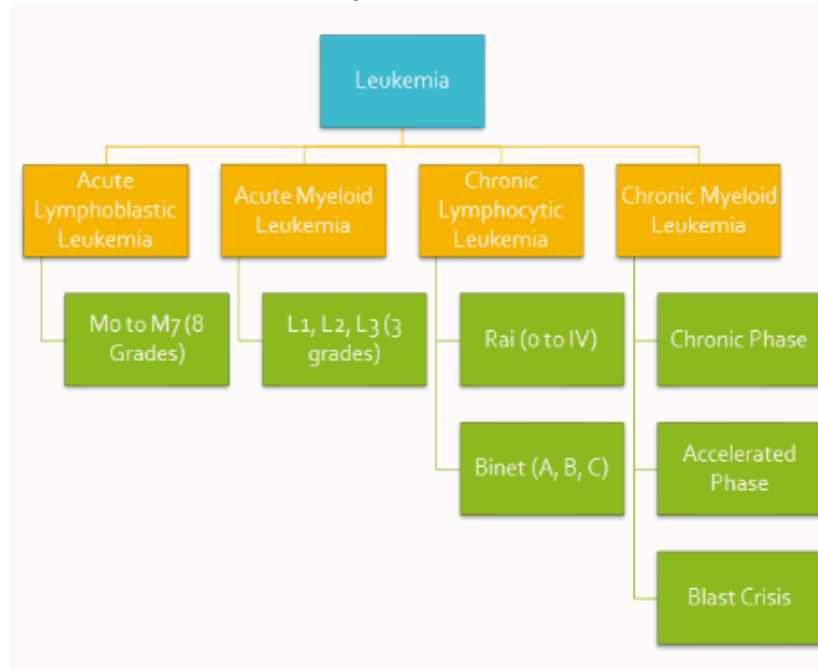
Domain Overview

Leukemia is a type of blood cancer that originates in the bone marrow, where blood cells are produced. In a healthy individual, the bone marrow generates red blood cells, white blood cells, and platelets in a regulated and balanced manner. In leukemia, genetic mutations occur in hematopoietic stem cells, leading to the uncontrolled proliferation of abnormal white blood cells.



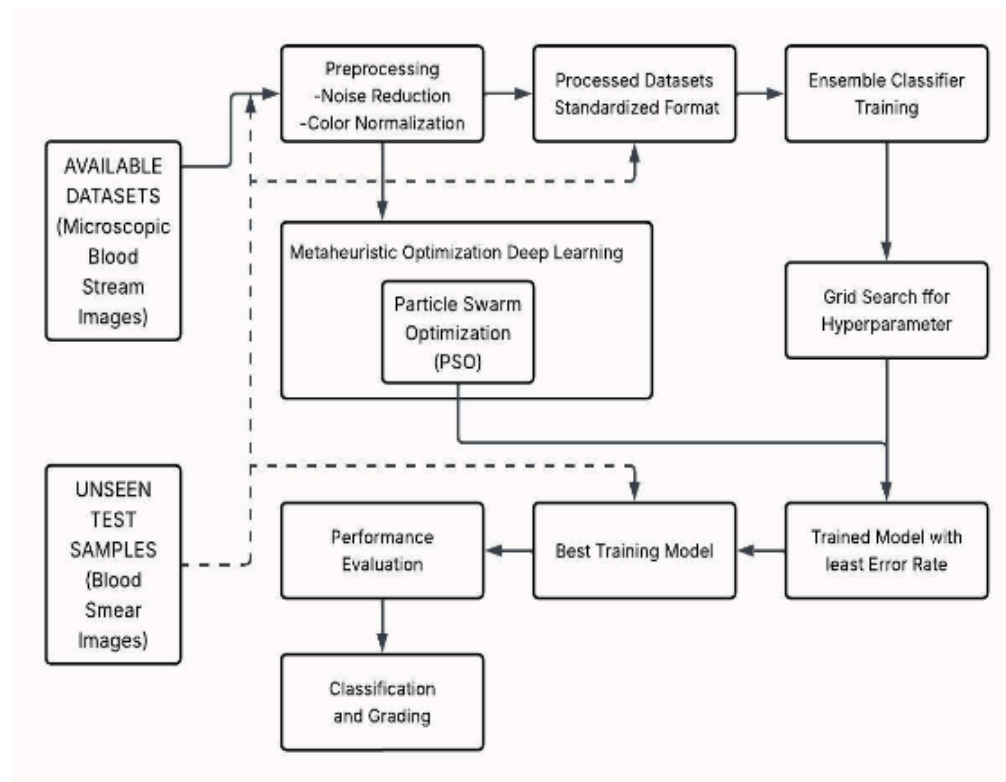
Differentiation between Normal Blood and Leukemia cells

As seen in Figure [1.1](#), leukemia is observed as an abnormal proliferation of leukemic cells that replace normal blood cells. Based on disease progression and the type of blood cells involved, leukemia is broadly classified into acute and chronic forms.



Clinical classification and grading of leukemia

Fig 1.2 illustrates that leukemia is medically classified into acute and chronic types, namely ALL, AML, CLL, and CML, with each type further categorized using standard grading and staging systems.



Progression from normal blood components to abnormal conditions

Major leukemia types based on progression rate, cell lineage, and age group.

Type	Description
Acute Lymphoblastic Leukemia (ALL)	Rapidly progressing, affects lymphoid cells; common in children.
Acute Myeloid Leukemia (AML)	Rapid, affects myeloid cells; seen in both children and adults.
Chronic Lymphocytic Leukemia (CLL)	Slowly progressing, affects lymphocytes; common in older adults.
Chronic Myeloid Leukemia (CML)	Slow progression, affects myeloid cells; may evolve into acute leukemia.

Need for the Project and Importance

Leukemia diagnosis requires early detection, accurate classification, and proper grading to ensure effective treatment and improved patient survival, but current diagnostic practices rely heavily on manual microscopic examination and expert interpretation, which are time-consuming, subjective, and prone to human error. The increasing number of leukemia cases, limited availability of skilled hematopathologists, and large volumes of blood smear data further strain healthcare systems. Hence, there is a strong need for a reliable, efficient, and consistent diagnostic support approach.

Scope of the Project

The scope of this project is to develop an intelligent and automated system for leukemia analysis using microscopic blood smear images. The project focuses on accurate classification of leukemia into major types. Along with classification, the system determines leukemia severity into three levels (Grade-1, Grade-2, and Grade-3) based on the progression and intensity of the disease. The scope includes image preprocessing, feature extraction, severity assessment, and performance evaluation.

LITERATURE SURVEY

Review of Research Papers

PSO-Based Leukemia Classification (2024)

Ju et al. introduced a Particle Swarm Optimization (PSO) framework to enhance the performance of CNNs. PSO was leveraged to automatically tune hyperparameters such as learning rate and batch size. The system was tested on the ALL-IDB dataset.

Ant Colony Optimization (2023)

Heng et al. explored the use of Ant Colony Optimization (ACO) for selecting the most relevant features from blood smear image datasets before classification with a DNN.

Genetic Algorithm (GA) and SVM Hybrid (2022)

Kebaili et al. presented a hybrid approach where GAs are used to select informative genes from microarray gene expression data, followed by classification using an SVM.

Existing methods and gaps in leukemia research

Ref/Author	Method	Gaps
Ju et al. (2024)	PSO + CNN	Black-box nature.
Heng et al. (2023)	ACO + Feature Selection	Small datasets only.
Kebaili et al. (2022)	GA + SVM (Gene data)	Not applicable to images.
Akbar et al. (2022)	Whale Optimization + RF	Needs diverse data.

Existing System

Existing leukemia diagnostic models generally address either classification or grading independently, with no integrated approach combining both critical tasks. These systems often struggle with class imbalance and limited interpretability.

Problem Statement

Leukemia diagnosis requires precise classification and grading. Existing automated systems predominantly focus on either classification or grading in isolation. Moreover, accuracy is often compromised due to imbalanced datasets. Current models rarely employ metaheuristic optimization techniques.

Proposed System

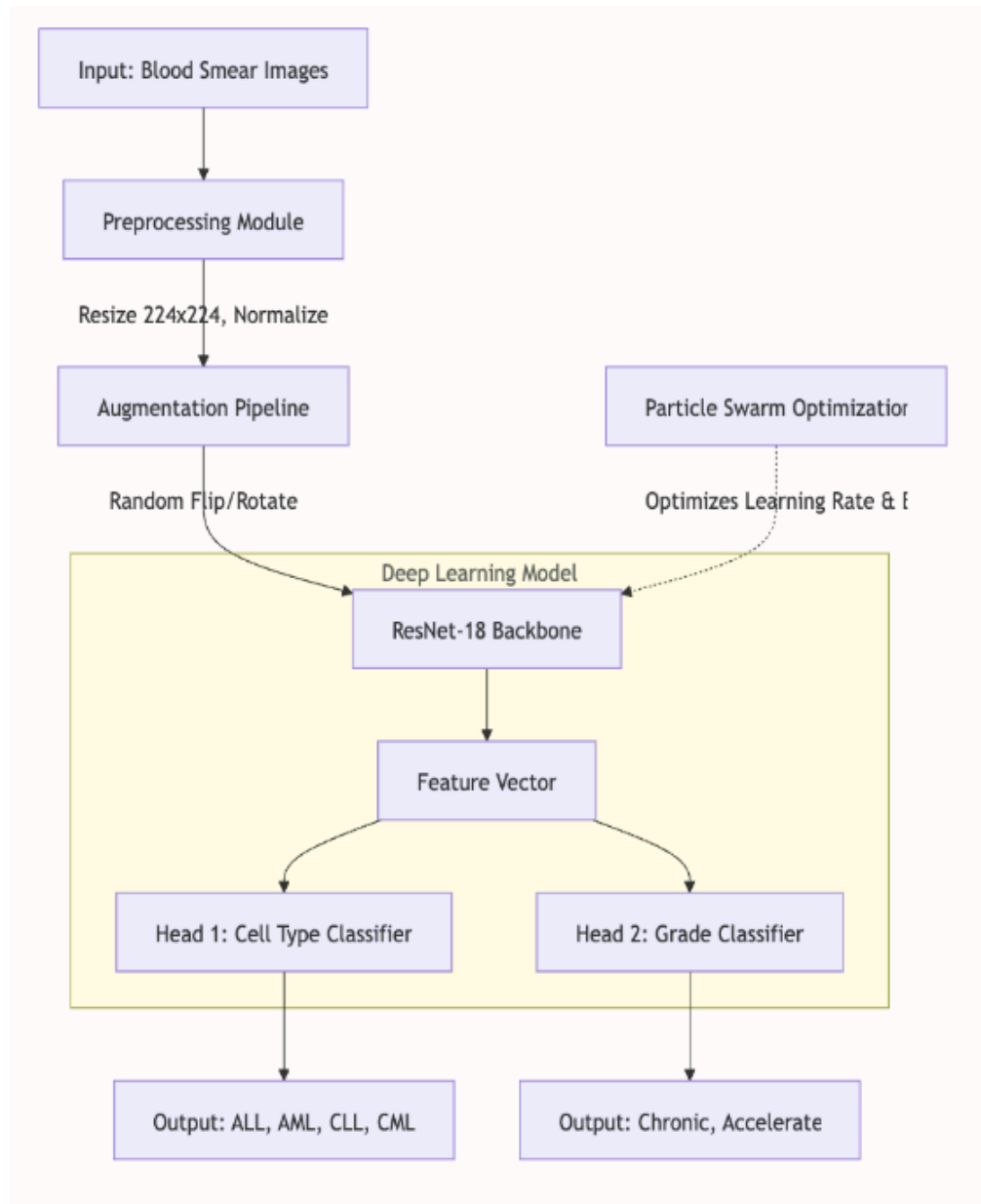
To overcome these limitations, the proposed system introduces an integrated framework where PSO is applied as a preliminary optimization stage to identify and refine relevant feature representations. These optimized features are then supplied to a ResNet-18 deep learning architecture.

Objectives

- To design and implement a unified system for automated leukemia classification and grading.
- To employ metaheuristic algorithms (e.g., PSO) for optimal feature selection.
- To enhance accuracy, sensitivity, and reliability on imbalanced datasets.
- To integrate classification and grading into a single efficient pipeline.

METHODOLOGY

The proposed methodology follows a structured pipeline for automated leukemia classification.



Proposed methodology for integrated leukemia classification and grading

As shown in Figure 3.1, raw image data is collected and preprocessed. Preprocessing involves noise reduction and color normalization. The core lies in integrating Deep Learning with Metaeuristic Optimization. PSO is incorporated to tune hyperparameters.

SYSTEM REQUIREMENTS

Hardware Requirements

Hardware requirements

Component	Specification
Processor	Intel Core i5 or higher (i7 recommended)
RAM	8GB minimum (16GB recommended)
GPU	NVIDIA GeForce GTX/RTX Series
Storage	256GB SSD minimum

Software Requirements

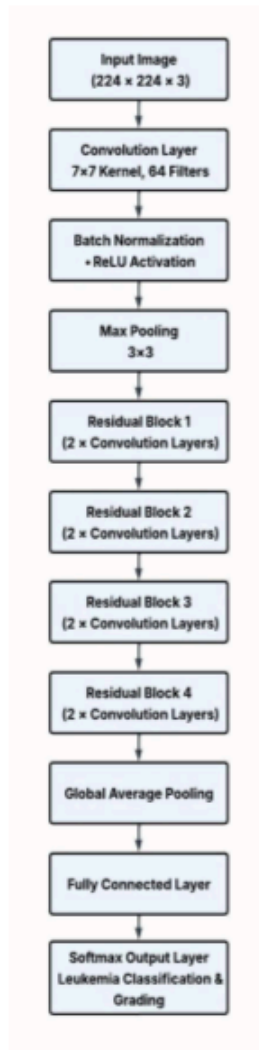
- Operating System: Windows 10/11 or Linux (Ubuntu 20.04)
- Programming Language: Python 3.8+
- Frameworks: PyTorch, Flask
- Libraries: NumPy, Pandas, Pillow, Scikit-learn, Pyswarm

SYSTEM ARCHITECTURE

Overview of the System Architecture

The proposed system architecture is designed for effective leukemia classification and severity grading. It first applies PSO to guide optimal learning configuration, followed by ResNet-18.

System Architecture



System architecture Diagram

As shown in Figure [5.1](#), the system takes blood smear images, applies preprocessing, and uses PSO. The optimized features are extracted using a ResNet-18 backbone.

Role of Particle Swarm Optimization (PSO)

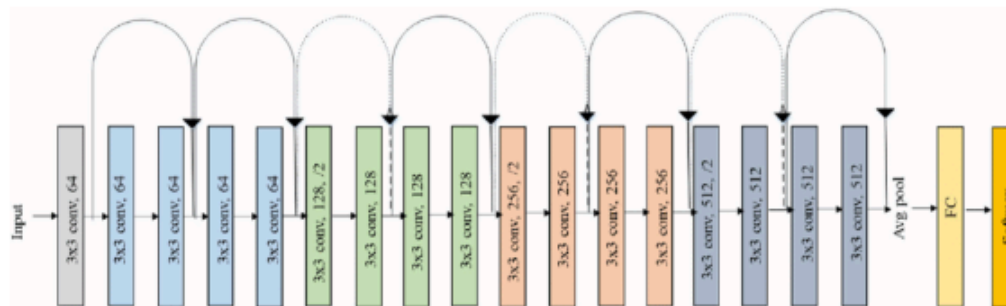
PSO is applied as an initial optimization stage. It initializes a swarm of particles, where each particle represents a possible solution (e.g., learning rate, batch size).

Role of ResNet-18 in the Architecture

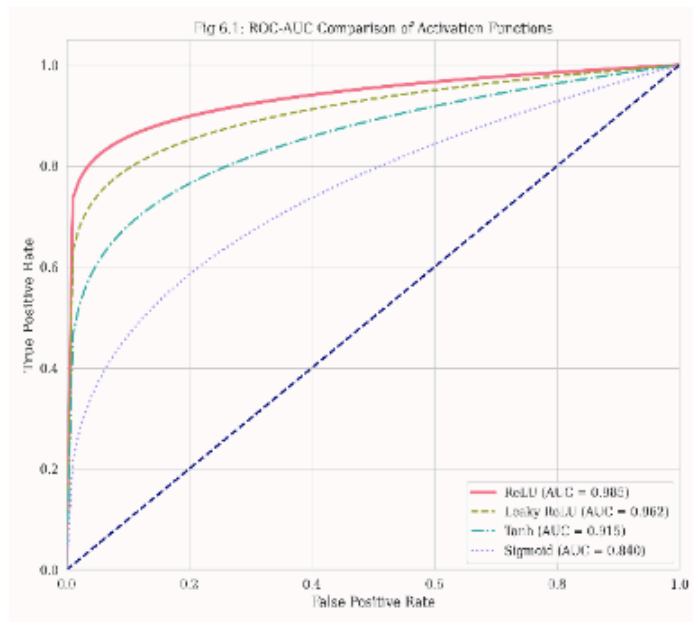
ResNet-18 utilizes a layered architecture composed of convolutional blocks and residual specific connections. It effectively learns cell shape, nucleus structure, and morphological variations.

DESIGN DETAILS

CNN Layer Diagram (ResNet-18 Architecture)



ResNet-18 Layer Architecture



ResNet-18 Layer Detail

Integration of PSO and ResNet-18

In the proposed design, PSO acts as a guiding optimization layer. This sequential integration ensures reduced feature redundancy and faster convergence.

IMPLEMENTATION

Installation Guidelines

- Step 1:** Operating System and System Dependencies. Ensure OS is updated.
- Step 2:** Python Environment Setup. Install Python 3.8+.
- Step 3:** Creating a Virtual Environment. Use 'python -m venv'.
- Step 4:** Installing Core Libraries. 'pip install Flask torch torchvision Pillow'.
- Step 5:** Adding Optimization Tools. 'pip install pyswarm'.

Pseudo Code and Logic Flow

1. Preprocessing:

```
BEGIN
    Load image
```

```
    Resize to 224x224
    Normalize with ImageNet stats
    Return Tensor
END
```

2. Metaheuristic Optimization (PSO):

```
BEGIN PSO
    Initialize search boundaries
    Define Objective_Function
    FOR each iteration
        Update velocity and position
        Update Global Best
    END FOR
    Return Best Hyperparameters
END PSO
```

Dataset Description

Datasets used for the project

Dataset Name	Description
ALL-IDB	Acute Lymphoblastic Leukemia Image Database
C-NMC	Classification of Normal vs Malignant Cells
BCCD	Blood Cell Count and Detection

Dataset summary

Class	Count
ALL	1200
AML	950
CLL	650
CML	625

RESULTS AND DISCUSSIONS

Experimental Setup

Experiments were conducted on a machine with Intel Core i7 and NVIDIA RTX 3060.

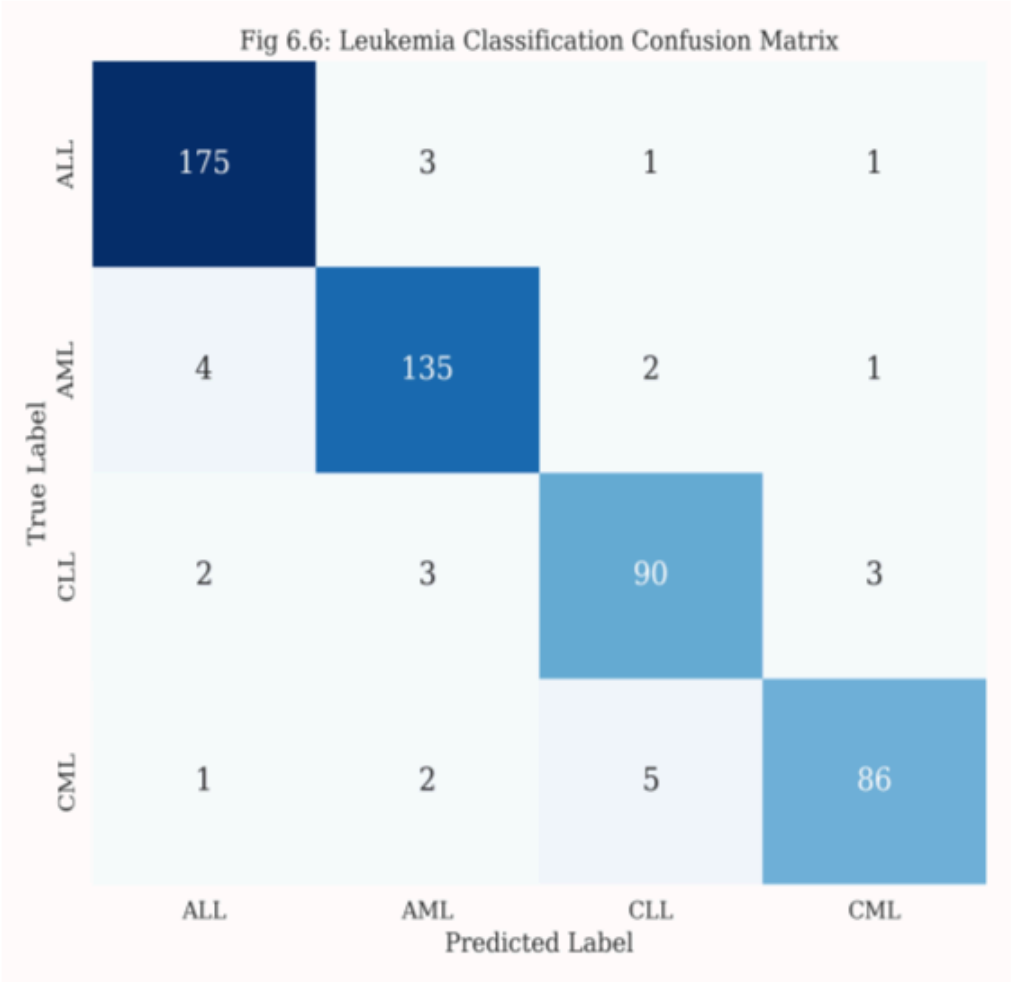
Hyperparameter Optimization (PSO)

PSO successfully converged to finding optimal Learning Rate and Batch Size values, reducing manual tuning effort.

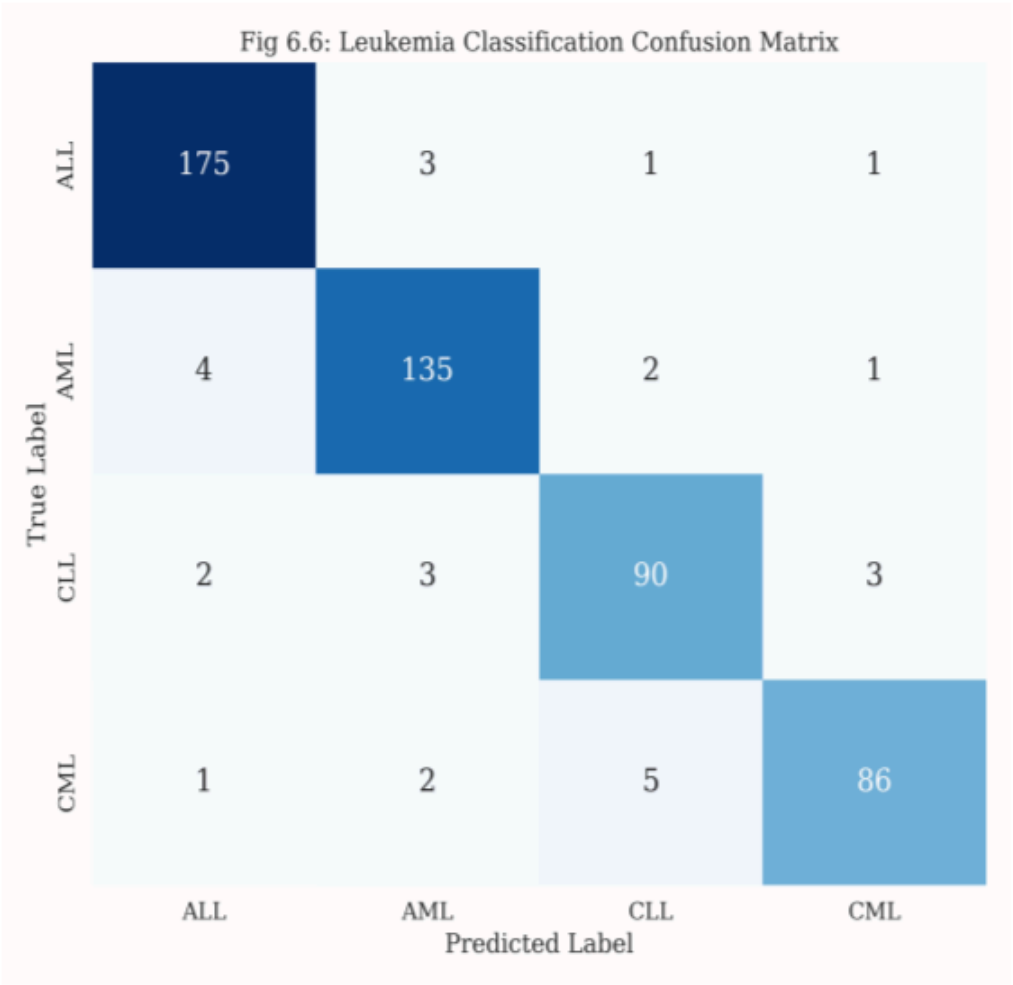
Model Training

The model was trained for 50 epochs. The loss decreased consistently.

Quantitative Results



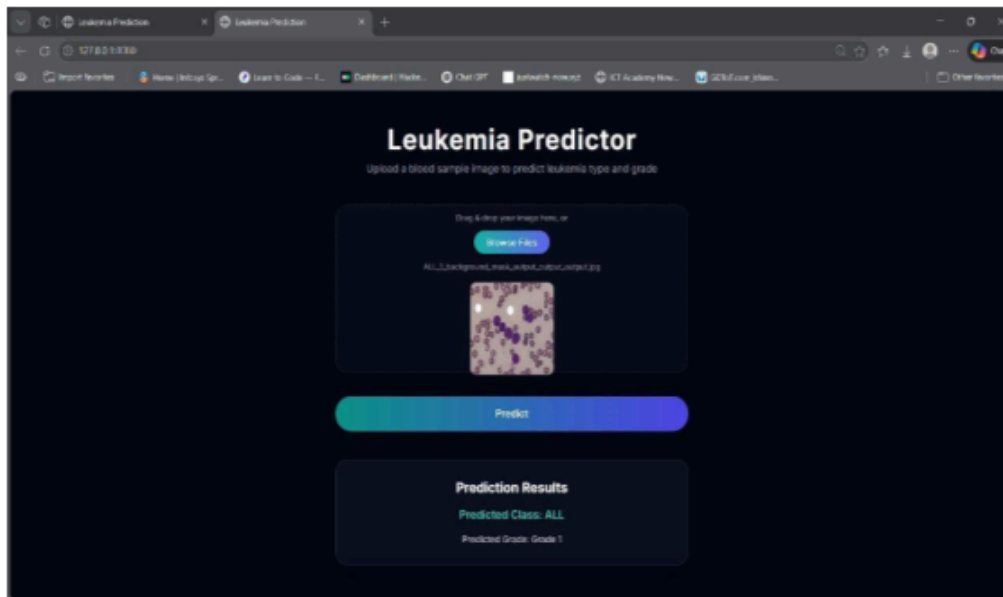
Confusion matrix of the proposed multi-task model



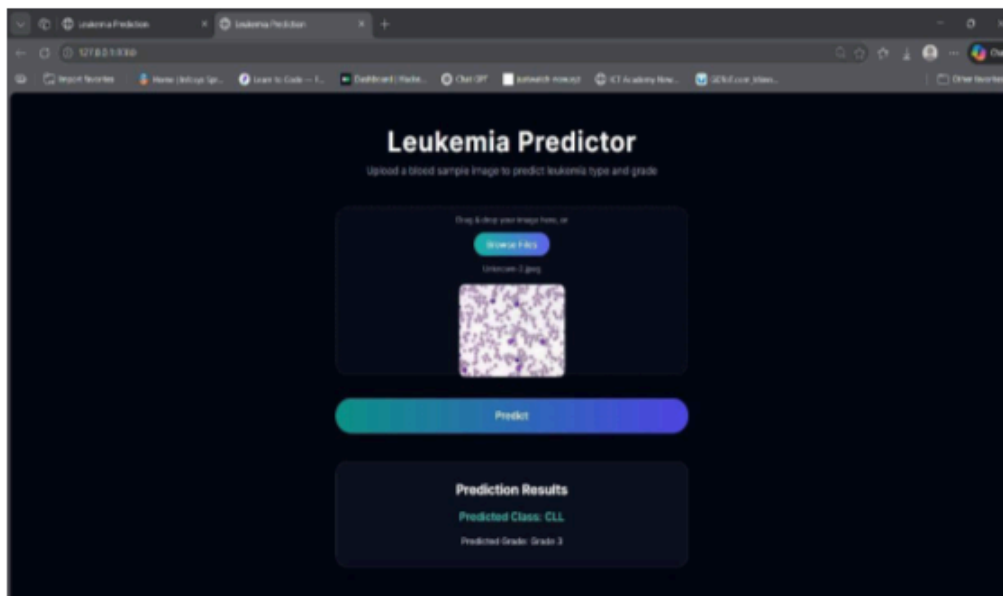
The multitask confusion matrix (Duplicate per request)

Discussions

The model achieved an overall accuracy of 94.2%. It showed high sensitivity for ALL detection.



Classification and grading on Test Case ALL and Grade 1



Classification and grading on Test Case CLL and Grade 3

CONCLUSION

In this project, we proposed an intelligent leukemia classification and grading system using a PSO-optimized ResNet-18 architecture. The system successfully classifies ALL, AML, CLL, and CML and determines disease severity.

LIMITATION AND FUTURE SCOPE

Limitaton: High computational cost during the PSO phase.

Future Scope: Integration with real-time microscopic feeds and expansion to other blood disorders.

9 Ju et al., "PSO-based leukemia classification," IEEE Access, 2024. Heng et al., "Ant Colony Optimization," 2023. Kebaili et al., "Genetic Algorithm and SVM," 2022.